

ACRME

Azure Capacity Reservation Management Engine

Key Architectural Diagrams

Presentation & Infographic Reference

The Six Core Architecture Views

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System Architecture Overview

End-to-end control plane spanning Prod, NonProd and DR regions

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Two quota groups per region and why the NonProd+DR group is Tier-3 quota-neutral

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Tier-3 execution: safe Path B (VM keeps running) with Path A deallocate fallback

System Architecture Overview

End-to-end control plane spanning Prod, NonProd and DR regions

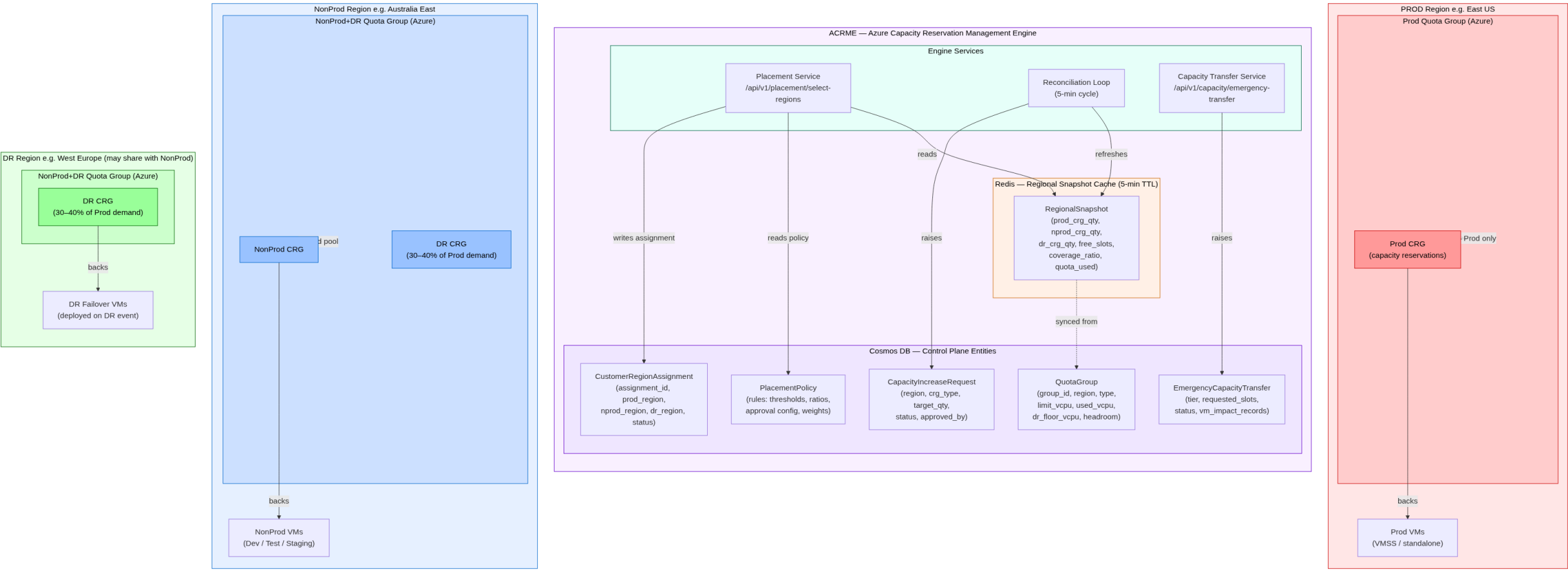
3 isolated regions: Prod CRG | NonProd CRG | DR CRG (30-40% of Prod demand)

Engine services: Placement, Reconciliation (5-min), Capacity Transfer

Redis RegionalSnapshot cache (5-min TTL) synced from Cosmos DB control plane

Cosmos DB entities: Assignment, Policy, IncreaseRequest, QuotaGroup, Transfer

Diagram 01 — System Architecture Overview



Quota Group Architecture

Two quota groups per region and why the NonProd+DR group is Tier-3 quota-neutral

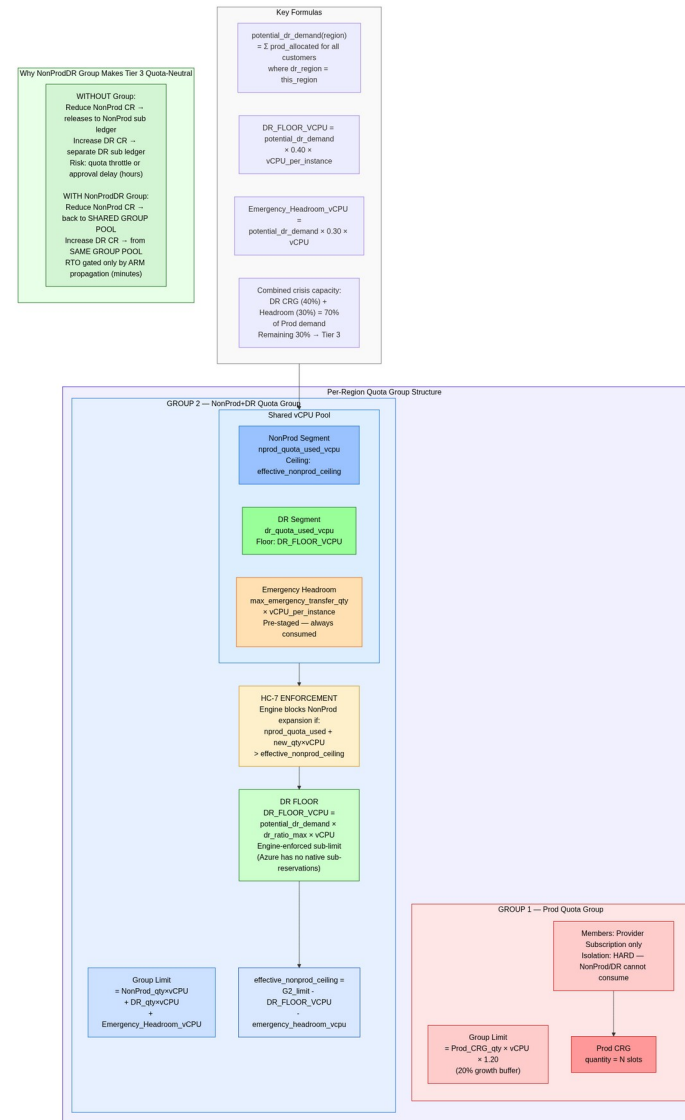
Group 1 — Prod: HARD isolation, NonProd/DR cannot consume; +20% growth buffer

Group 2 — NonProd+DR: shared vCPU pool (NonProd / DR / Emergency headroom)

DR floor & emergency headroom are engine-enforced sub-limits (no native Azure sub-reservations)

Shared pool → Tier-3 transfers are quota-neutral; RTO gated only by ARM propagation

Diagram 02 — Quota Group Architecture



Placement Engine Flow

How a placement request is filtered through 7 hard constraints and scored

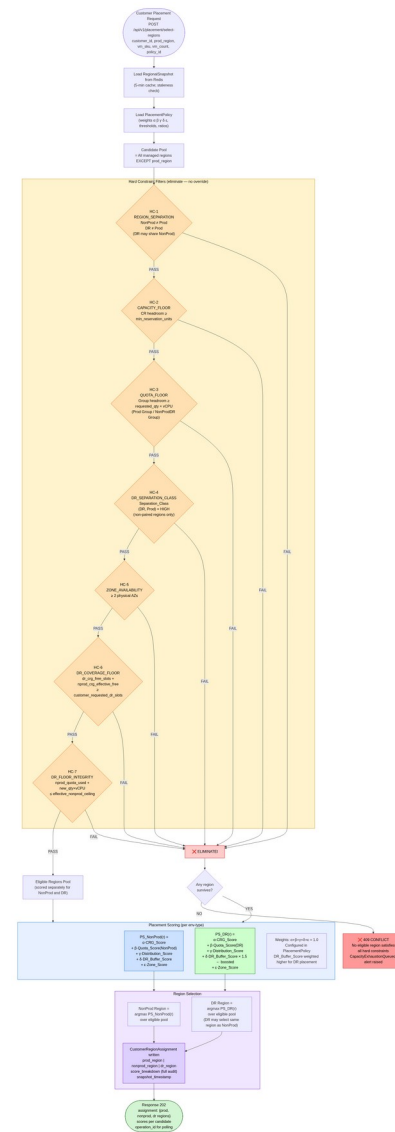
Loads RegionalSnapshot + PlacementPolicy; candidate pool excludes prod_region

HC-1..HC-7 hard filters eliminate regions (no override possible)

Weighted scoring $PS = \alpha \cdot CRG + \beta \cdot Quota + \gamma \cdot Dist + \delta \cdot DR_Buffer + \epsilon \cdot Zone$

Selects NonProd & DR regions independently; writes audited CustomerRegionAssignment

Diagram 03 — Placement Engine Flow



Steady-State Capacity Lifecycle

Auto-increase triggers, debounce guard and approval-gated vs self-managed phases

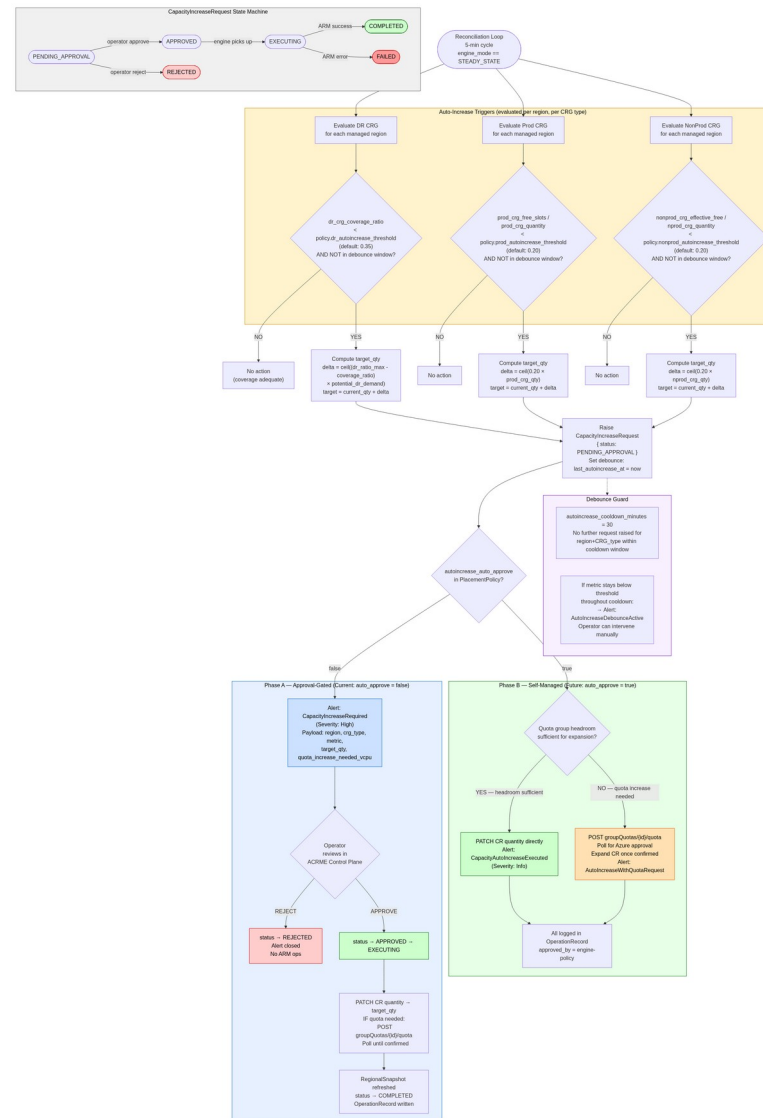
Reconciliation loop evaluates DR / Prod / NonProd CRGs against auto-increase thresholds

DR threshold 0.35, Prod/NonProd 0.20; 30-min debounce cooldown per region+CRG type

Phase A (current): approval-gated operator review in ACRME Control Plane

Phase B (future): self-managed auto-approve when quota headroom is sufficient

Diagram 04 — Steady-State Capacity Lifecycle



Emergency Capacity Transfer

Three-tier crisis response, only valid when `engine_mode == DR_EVENT_ACTIVE`

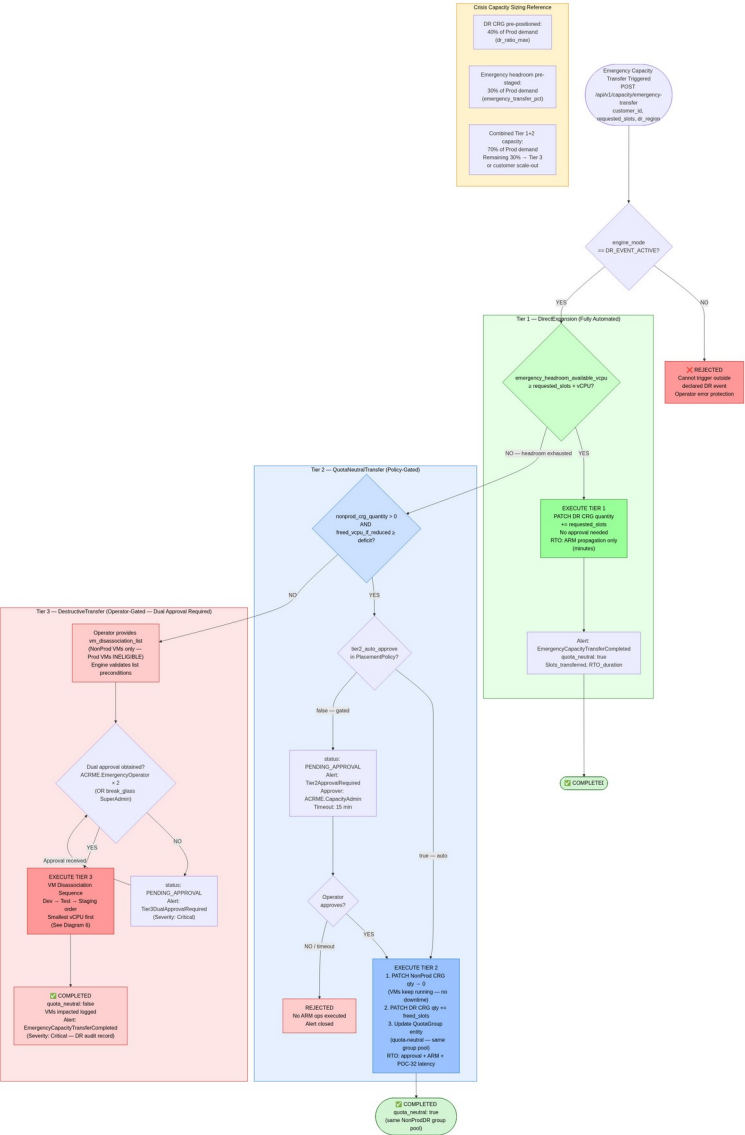
Tier 1 DirectExpansion: fully automated from pre-staged headroom (RTO = ARM minutes)

Tier 2 QuotaNeutralTransfer: policy-gated, reduce NonProd → expand DR, no downtime

Tier 3 DestructiveTransfer: operator dual-approval; VM disassociation (see Diagram 6)

Sizing: DR 40% + headroom 30% = 70% of Prod demand; remaining 30% → Tier 3

Diagram 05 — Emergency Capacity Transfer



VM Disassociation Sequence

Tier-3 execution: safe Path B (VM keeps running) with Path A deallocate fallback

Precondition validation: NonProd-only VMs, CRG association & coverage confirmed

Order: Dev → Test → Staging; smallest vCPU first

Path B (default): reduce provider CR → clear VM CRG ref → VM continues running

Path A fallback: deallocate VM (restart required); every step logged as VM_ImpactRecord

Architecture at a Glance

- Hard region isolation: Prod $\neq$ NonProd $\neq$ DR, enforced across placement & quota
- Quota-neutral crisis response via shared NonProd+DR pool (Tiers 1-3)
- Deterministic, fully-audited placement through 7 hard constraints + weighted scoring
- Steady-state auto-increase with debounce; DR-safe VM disassociation ordering